School of Computer Science and Engineering

BCSE301L-Software Engineering

Tutorial – 5

Faculty: Dr. S.Ebenezer Juliet **Winter Semester 2025-2026**

Class Id: VL2025260503715 **Slot:A2+TA2**

Date: 11-03-2026

1	a)A university develops a Student Information System. A module named StudentServices performs the following operations: Student registration Fee payment processing Hostel room allocation Library book issue **Answer the following** i. Identify the **type of cohesion present** in this module. ii. Explain why this cohesion is considered weak or strong. iii. Redesign the module structure to improve software quality. b) A software project contains the following modules: • Module A sends the entire customer record to Module B. • Module B only uses the customer ID to process the request. **Answer the following** i. Identify **the type of coupling present**. ii. Explain why this type of coupling is undesirable. iii. Suggest a better design approach to reduce coupling.
2	A banking application is designed with the following components: • User interface for customers to access banking services • Business logic that processes transactions and validates operations • Data access layer that interacts with the database • Database layer storing customer and transaction records Each layer communicates only with the layer directly below it. i. **Identify the architectural style used** in this system and justify ii. Explain the role of each layer in the architecture. iii. Discuss the advantages of using this architecture in terms of maintainability and scalability. b) A software development tool includes modules for code editing, compiling, debugging, and version control. All modules share and update a central repository containing the project source code and metadata. i. Explain **how repository architecture** is applied in this scenario. ii. Identify the benefits of having a central repository.

	c) An online shopping website allows users to search for products, view product details, and place orders. The system separates the application into three components: • One component manages the data and business logic • One component handles user interface display • Another component processes user input and interactions Answer the following i. **Identify the architectural pattern used** in this scenario. ii. Explain the roles of **Model, View, and Controller** in this system. iii. Discuss how this architecture improves modularity and maintainability.
3	A food delivery application includes entities such as Customer, Restaurant, Menu, Order, and Payment. **Apply object-oriented design principles.**
4	A mobile health monitoring application displays real-time data such as heart rate, step count, and calorie consumption. **Apply the steps of user interface design process**. a) Explain the user interface design guidelines that should be followed for mobile devices. b) Suggest how the interface can display real-time data effectively. c) Explain how the design can support different types of users (elderly, athletes, patients)
5	An online shopping system allows a customer to purchase products through the website. The process involves the following steps: 1. Customer logs into the system. 2. Searches and selects products. 3. Adds products to the cart. 4. Proceeds to checkout. 5. System verifies payment details. 6. If payment is successful, the order is confirmed and a receipt is generated. 7. If payment fails, the system asks the customer to retry payment. Draw a UML **Sequence diagram/Activity Diagram / State transition diagram** representing the workflow of this system**. [Remember the notations/ symbols of each behavioural and functional model]**

6	Consider the following program segment
	1.Void Selection Sort (int array1[]) {
	2. int size=array1.length;
	3. for (int i=0; i<size; i++){
	4. int minIndex=i;
	5. for (int j=i+1; j<size; j++) {
	6. if(array1[j] < array1[minIndex])
	7. minIndex=j; }
	8. int temp= array1[minIndex];
	9. array1[minIndex] =array1[i];
	10. array1[i] =temp;
	11 .}
	12. }
	a) Prepare the **CFG and determine the cyclomatic complexity of the selection sort** function.
	b) Design **a test suite for the selection sort that satisfies the Path coverage** (Show the important steps in your test suite design method).

7	b) You are assigned with testing a banking application's transfer functionality. The system allows users to transfer funds between their own accounts and to other accounts within the same bank. **Perform Black box testing method** to ensure the transfer functionality is working correctly in the bank. **(Boundary value analysis and Equivalence Partitioning)** (6 Marks) • The system allows users to transfer funds within the same bank only, not to accounts in other banks. • Users can transfer funds between their own accounts or to other accounts within the same bank. • The transfer functionality includes entering the recipient's account number, the amount to transfer, and any optional message. • There are two types of accounts in the bank: savings account and checking account.
8	a)You have been appointed as a purchase officer of the beauty soap manufacturing company and your job is to maintain the production plant in such a way that there is no interruption in the production process for 24x7. In order to increase the production rate to match the demand in the market, your company decided to procure an equipment for its production plant. As a purchase officer you have to undergo a series of process to complete the equipment procurement process as per the company policies. The company policies can be assumed as per your wish. **Demonstrate the data flow diagram in all the three levels with proper precisions.** 3b. Construct a **structured design to provide convenient transition from data flow diagram to software architecture.** Follow all the recommended steps for the architectural mapping in order to provide an appropriate program structure for the above-mentioned scenario. **(refer8a)**

9	a). You are provided with a simplified JavaScript function for a web-based application that calculates and displays the yearly subscription cost for a user based on the number of users and the chosen subscription plan. The application offers two plans: 'basic' at $100 per user per year and 'premium' at $150 per year. A discount is also applied for more than 100 users: a 10% discount for the 'basic' plan and a 15% discount for the 'premium' plan.
	function calculateYearlySubscriptionCost(numberOfUsers, planType)
	{
	let costPerUser = planType = 'basic' ? 100: 150;
	let discount = numberOfUsers > 100 ? (planType = 'basic' ? 0.1: 0.15) : 0;
	return numberOfUsers * costPerUser * (1 - discount);
	}
	Calculate the **cyclomatic complexity** for the above-mentioned routine appropriately.

	b) **Design the test cases** in such a way that it **covers all the paths of the given routine**. Every testcase should brief the rationale to specify its necessity. 1. Test with the 'basic' planType more than 100 users 2. Test with the 'basic' planType less than 100 users 3. Test with the 'premium' planType more than 100 users 4. Test with the 'premium' planType less than 100 users Totally, 4 test cases to achieve full branch coverage
10	a) A software company has completed the Software Requirement Specification (SRS) for a new online banking system. Before starting development, the project manager arranges a verification meeting where developers, testers, and domain experts read and

	discuss the document to identify missing requirements, inconsistencies, and unclear statements, and to suggest improvements to enhance the quality of the requirements. **Answer the following:** a) Identify the **type of verification activity** used in this scenario. b) Explain the outcome of this verification activity. b)A development team has implemented a payment processing module. Before integrating the module into the system, **a structured meeting is organized where team members examine the source code line by line to detect logic errors**, coding standard violations, and security vulnerabilities. Answer the Question: i)**Identify the verification technique applied in this scenario**. ii)Explain how this technique differs from a simple review. iii) Discuss the advantages of using this technique in software quality assurance. c) A software company is developing a **hospital management system**. The organization follows standard quality procedures for requirement analysis, design, coding, and testing. Before releasing the product, **an independent quality team examines whether the project activities follow the defined standards and processes**. Question: i) Identify the **type of quality activity performed in this scenario**. ii)Explain the purpose of software auditing. iii) Describe the steps involved in conducting a software audit.
11	Consider a mobile application for understanding the various events happening in our university. This application should list the events in different ways based on event date, event hosting department and event category (both technical and non-technical). It should support event registration, cancellation, payment and rating. For the given scenario develop the following a) A technical diagram to **show the attributes and methods of every entity involved**. **Class diagram** should be drawn with different type of attributes, methods, aggregation, composition and cardinality ratio. b) A technical diagram to show the object interactions. **Sequence diagram:** Object interactions, looping and numbering should be shown properly.
12	You are part of a software development team tasked with designing an air traffic control (ATC) system, a hard real-time system where delays or failures could lead to catastrophic consequences. The system is responsible for monitoring and managing aircraft movements within a designated airspace, ensuring safe distances between planes, guiding them during take-off and landing, and providing real-time alerts to air traffic controllers. Radar systems continuously track the position, altitude, speed, and heading of all aircraft in the airspace and feed this data into the ATC system.. The system processes the radar data and displays it on a graphical interface for air traffic controllers, enabling them to make informed decisions. It also calculates potential collision risks based on predefined safety thresholds, such as minimum separation distances. If a potential collision is detected, the system immediately generates an alert and recommends corrective actions, such as adjusting altitude or heading.. Instructions are then communicated to pilots via radio or automated messaging systems. To ensure uninterrupted operation, the system must seamlessly switch to a backup server within

	milliseconds in the event of hardware or network failure, maintaining the integrity and reliability of the entire operation. i For this scenario, **model the structural blueprint of the system**, outlining its key entities and their interconnections using UML approach. ii Depict the **temporal ordering of method calls/messages and object interactions** within the system using UML approach.
13	Below are six modules interaction given for a Library Management system. Identify the type of **coupling in the interaction and types of cohesion** that the modules exhibit wherever applicable. Justify the answer. i When a user logs in, the User Management Module verifies the credentials and checks for any pending fines. It then sends the user's status (active/inactive) and fine details to the Borrowing Module, which decides whether the user can borrow a book. ii When a user borrows a book, the Borrowing Module directly accesses and updates the book's availability status in the Book Management Module's database tables. iii When a book is borrowed, the Borrowing Module calls the Notification Module and passes the user's contact details, book title, and return due date as function arguments. The Notification Module then sends an email or SMS. iv The Report Generation Module retrieves structured data (e.g., JSON responses) from the Borrowing Module, Book Management Module, and User Management Module to generate a report on overdue books and most borrowed books. It does not modify any data, only reads it. v)Inside the Borrowing Module, the following steps occur sequentially when a user borrows a book: Validate user eligibility Check book availability Update borrowing records Adjust book availability Notify the user The Notification Module checks a user's preferred mode of communication (email/SMS/push notifications). It then selects a function based on this preference using multiple if-else conditions.
14	a) A Food Delivery System follows the workflow given below: A customer places an order by selecting items from a menu. The system verifies the payment and generates an order confirmation. The order is sent to the restaurant for preparation. Once prepared, a delivery agent is assigned. The delivery agent picks up the order and delivers it to the customer. Does this **system exhibit transaction flow or transform flow** ? Justify your answer by explaining how the control of execution flows through the different processes
	b) A payment system is being designed for an e-commerce website. There are two different ways to structure the system - The first method creates a general Payment class, with specific types like Credit_Card, PayPal, and UPI as subclasses. The second method starts with a single Payment_Processor module, which is later divided into separate handlers for each payment method. Which method uses abstraction, and which one uses refinement? Which approach is better for adding new payment methods in the future? Justify your answer.
15	For the code snippet provided below: i Calculate the cyclomatic complexity (4 Marks) ii Identify all independent paths (2 Marks) iii Write test cases to cover all independent paths (4 Marks) *#include <stdio.h>* *#include <string.h>* *float calculate_discount(float total_amount, int is_member, char coupon_code[]) {*

```
if (total_amount <= 0)
{
return 0;
}
float discount = 0;
if (is_member)
{
discount += total_amount * 0.1;
}
if (strcmp(coupon_code, "SAVE20") == 0)
{
discount += total_amount * 0.2;
} else if (strcmp(coupon_code, "SAVE10") == 0)
{
discount += total_amount * 0.1;
}
float max_discount = total_amount * 0.3;
if (discount > max_discount)
{
discount = max_discount;
}
float final_amount = total_amount - discount;
return final_amount;
}
```